Sharp thresholds for five-colour cubes

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Abstract

Every collection of fifteen cubes, each displaying all five colours of a fixed palette, contains eight that form a $2 \times 2 \times 2$ cube with monochromatic exterior faces using all five colours. Fourteen need not suffice. This settles the five-colour corner value listed as unknown by Berkove, Eickemeyer and Schulman (2018). An elementary Hall argument then shows that any n^3 such cubes can all form a complete $n \times n \times n$ cube for $n \geq 3$, and that $12n - 16$ cubes suffice for its frame for every $n \geq 3$. With at most k source varieties, the sharp corner threshold is 8 for $k = 1$, 13 for $k = 2$, and 15 for $k \geq 3$. The corner lower bound is elementary; the upper bound uses 279 finite interval certificates. Lean verifies these certificates, their connection to physical arrangements, the sharp frame and complete-cube thresholds, and the bounded-variety thresholds.

AI transparency. The research, code and manuscript were developed using GPT-6-Astra (OpenAI), through Codex, under the author's direction. This working manuscript has not received independent human peer review; Section 8 states the formal verification scope.

1 Introduction

MacMahon's coloured-cube puzzles [1] ask when a collection of face-coloured cubes can form a larger cube with monochromatic exterior faces. Here every source and target uses all five colours of a fixed palette. Proper rotations identify varieties; reflections are not allowed as moves. Collections may contain repeated varieties, and touching interior faces need not have equal colours. The target may depend on the collection.

For $n \geq 2$, an *n-frame* consists of the corners and edges of an $n \times n \times n$ cube, using $12n - 16$ pieces. Write $F_5(n)$ for the least available pool size guaranteeing an n-frame, and $G_5(n)$ for the least pool size guaranteeing a complete n-cube. An assembly may use a subset of the pool. At $n = 2$, the frame and complete cube both consist of eight corners; we write $N_5 = F_5(2) = G_5(2)$.

Berkove, Eickemeyer and Schulman [2, Section 8, Table 7] tabulate frame numbers for palettes of two to six colours, leaving the five-colour column unknown. Their corner values for two, three, four and six colours are respectively $11, 23, 11, 24$. We determine the five-colour column.

Theorem 1. *The sharp five-colour corner threshold is $N_5 = 15$.*

The conclusion for larger collections follows by choosing any fifteen cubes. A short corner argument gives an explicit fourteen-cube obstruction. For the upper bound, Hall deficiencies and symmetry reduce hypothetical counterexamples to 357 finite cases. Containment, budget bounds and a deletion lemma leave 279 smaller problems, each excluded by an interval certificate (Sections 4–5).

The corner theorem also yields the exact functions

$$F_5(n) = \max\{12n - 16, 15\}, \qquad G_5(n) = \max\{n^3, 15\} \qquad (n \geq 2).$$

Thus any twenty cubes form a 3-frame, and any 27 form a complete 3-cube, without spares. Section 6 uses only the corner theorem, elementary geometry and Hall's theorem; it requires no further interval certificates. The key auxiliary lemma treats eleven cubes sharing a missing bichromatic edge type. Section 7 determines the thresholds with a bounded number of source varieties, showing in particular that a fourteen-cube obstruction needs at least three varieties. All these results are formalised in Lean; Section 8 describes the proof and its computational components. The accompanying review archive contains the certificates, programs, geometric audits and Lean sources.[1]

The companion paper [4] considers four-colour sources and targets using exactly three colours. It gives the sharp frame threshold $16n - 3$ for every side length $n \geq 2$, with a classification of the largest failing collections. The two developments share the palette-independent geometry in `ColourCubes` [5]; neither threshold theorem is used to prove the other.

A limited literature search has found no later value for the five-colour entry. This is not a comprehensive priority claim. In particular, Rosenfeld's October 2026 article [6] has not been examined in full.

2 Geometry and Hall deficiencies

Represent a cube by its colours on the faces $(+x, -x, +y, -y, +z, -z)$ in a right-handed coordinate frame. A five-colour cube has one repeated colour. There are 75 varieties: for each repeated colour, three have its two faces opposite and twelve have them adjacent. In the opposite case, the other four colours form two opposite pairs, giving three possibilities. In the adjacent case, fix the two equal faces; the 24 orders of the other colours are identified in pairs by the rotation exchanging those equal faces. Thus there are 15 opposite-repeat varieties and 60 adjacent-repeat varieties. Each class is one orbit under colour permutations.

A corner type is the ordered triple of its incident face colours, modulo cyclic permutation. Order the outward face normals positively, equivalently read the triple counterclockwise from outside along the body diagonal. This reverses the clockwise convention in [2], consistently on every cube, and therefore preserves all matching and containment relations. Repeated occurrences of a corner type are retained as distinct target positions.

Lemma 2 (Corner–rotation correspondence)**.** *A source cube fits a target corner by a proper rotation if and only if it has the same cyclic corner type at some corner.*

Proof. A proper rotation preserves the orientation of a corner frame, so it induces a cyclic permutation of its three outward faces. Conversely, any two positively oriented corner frames, with a prescribed cyclic alignment, are related by a determinant-positive signed permutation of the coordinate axes. This is a proper rotation of the cube. □

The palette-independent Lean version of this lemma is reused from the shared `ColourCubes` library [5].

Fix a target T. Form a bipartite graph whose left vertices are its eight corner positions and whose right vertices are the individual source copies. A copy is adjacent to a position when it fits there. A matching covering the eight positions is exactly a solution: each chosen copy can be rotated independently, since interior faces are unrestricted. By Hall's theorem [3], failure is equivalent to a nonempty set D of target positions having fewer than $|D|$ neighbouring copies.

Index the varieties by $i = 1, \ldots, 75$, and let x_i be their multiplicities. For a set S of varieties, write $x(S) = \sum_{i \in S} x_i$. Let $N_T(D)$ be the varieties fitting at least one position of D. Then failure of T is equivalent to the existence of a cut

$$x(N_T(D)) \leq |D| - 1, \qquad \varnothing \neq D \subseteq \{1, \ldots, 8\}. \tag{1}$$

[1] Intended repository: https://github.com/jasondavies/five-colour-cubes. Public release is pending; the review archive includes source checksums, the base Git revision and bundled pinned project dependencies.

There are 255 cuts per target, hence 19,125 cuts altogether.

3 An elementary fourteen-cube obstruction

Consider four copies of A, five of B, and five of C, where

$$A = (0, 0, 1, 2, 3, 4), \quad B = (0, 1, 2, 3, 2, 4), \quad C = (0, 3, 1, 2, 4, 1).$$

Their corner types are shown below. Triples denote cyclic classes; the last column counts positions whose type occurs in neither of the other varieties.

Variety	Corner types	Private positions
A	$013, 014, 023, 024, 031, 032, 041, 042$	5
B	$022, 023, 034, 042, 122, 124, 132, 143$	6
C	$011, 014, 021, 042, 113, 123, 134, 243$	6

Lemma 3. *The collection $4A + 5B + 5C$ has no five-colour solution.*

Proof. A solvable target must have every corner type in the displayed union. Its only two-colour types are $022, 122, 011, 113$. An adjacent-repeat target therefore repeats either colour 2, with types $022, 122$, or colour 1, with types $011, 113$. In either case these two corners determine the colours on the faces at the ends of the repeated-colour edge. The two orders of the remaining colours give a mirror pair: B and its mirror, or C and its mirror. The mirror of B requires 043, and the mirror of C requires 012; both types are absent. Only B, C remain.

An opposite-repeat target requires both cyclic orientations of each corner triple. The unordered triples with both orientations in the union are

$$013, 014, 023, 024, 123, 134.$$

If its repeated colour is r, delete r from the four triples containing it. The remaining pairs must form a $K_{2,2}$ on the other four colours: choose one face from each of their two opposite pairs. Colours 2 and 4 occur in fewer than four listed triples. For $r = 1$ or $r = 3$, the four pairs have degree sequence $(3, 2, 2, 1)$, so do not form $K_{2,2}$. For $r = 0$, the pairs are $13, 14, 23, 24$, forcing opposite pairs $\{1, 2\}$ and $\{3, 4\}$. This is A.

Thus only A, B, C can be targets. Their private positions require respectively at least five, six and six copies of the corresponding variety, exceeding the available four, five and five copies (Figure 1). None is solvable. □

4 Reduction to finite problems

Eight copies of any variety solve that variety, so a solution-free collection has every multiplicity at most seven. Suppose such a collection has size 15.

Lemma 4 (Anchored Hall cover)**.** *Suppose a collection contains m copies of a variety A and does not solve A. There is a set K of its target positions such that*

$$|K| \le 7 - m, \qquad \#\{\textit{other copies fitting some position outside } K\} \le 7 - m - |K|. \tag{2}$$

Conversely, such a set certifies failure of target A.

Proof. Apply (1) to A and put $K = D^c$. Every copy of A is a neighbour of the nonempty set D. Subtract these m copies from the bound $|D| - 1 = 7 - |K|$. The converse is the same Hall inequality read backwards. □

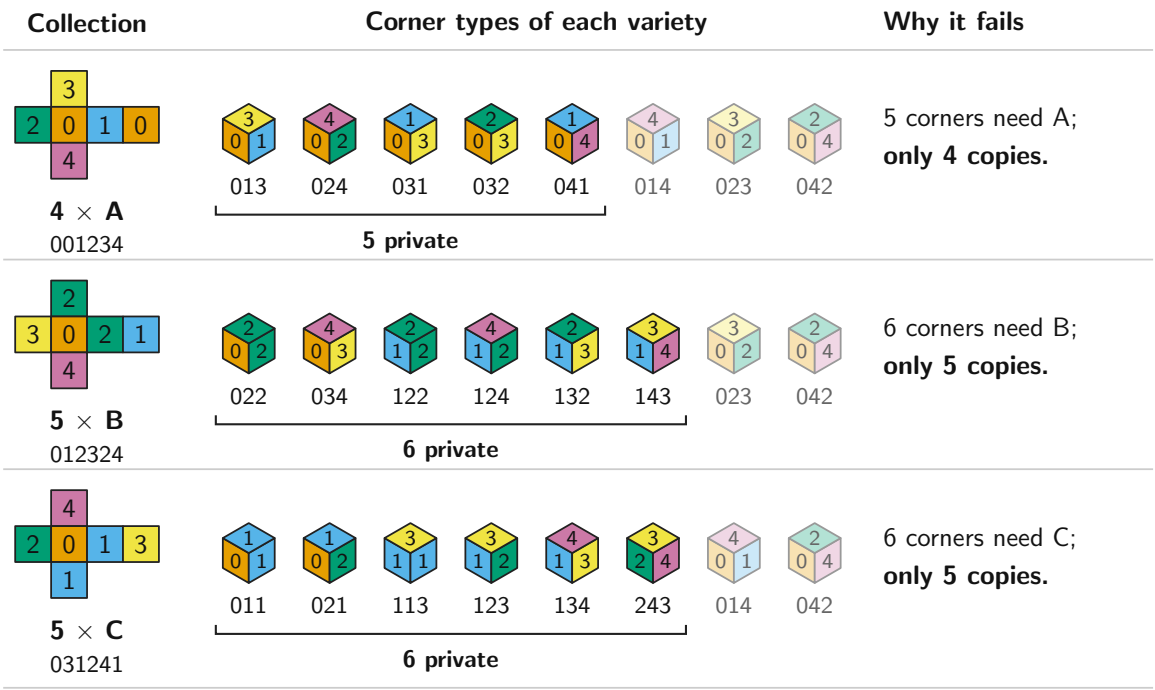


Figure 1: The fourteen-cube obstruction $4A + 5B + 5C$. Each net has $+x$ at its centre. Corner types are read counterclockwise from outside; the full-colour types occur on no other variety. A target of variety X has these private positions, and only copies of X fit them.

First suppose the maximum multiplicity m is at least two. Choose an adjacent-repeat anchor attaining m if possible; otherwise choose an opposite-repeat anchor attaining m. In the latter case all adjacent-repeat multiplicities are at most $m-1$. Relabel the colours to send the anchor to a fixed representative of its orbit.

For each cover K in Lemma 4, put

$$S(K) = N_A(K^c) \setminus \{A\}, \qquad b(K) = 7 - m - |K|.$$

Write $S = S(K)$ and $b = b(K)$. The case has $x_A = m$, $x(S) \le b$, and the multiplicity bounds just described. Additional upper bounds use two-variety solutions. If a copies of A and $8-a$ copies of V build a target, with $a \le m$, then $x_V \le 7 - a$ in a solution-free collection. These tests are finite: the two varieties must cover all eight target positions, A must fit at least a, and V at least $8-a$. They are also sufficient: at most a positions are unavailable to V, and all of them are available to A. Assign these to A, extend the assignment to a positions that A fits, and assign the remaining $8-a$ positions to V.

Colour permutations and simultaneous reflection of the entire collection preserve solvability. First identify equal pairs $(S(K), b(K))$, even when they arise from inequivalent position subsets K; then retain one representative per orbit of these pairs under the anchor stabiliser. The counts below refer to these constraint pairs. Transporting a cover also transports its constrained varieties, budget and multiplicities, so no counterexample is lost. Global reflection transports the whole collection; individual cubes still move only by proper rotations. The repeated-variety cases have the following counts, totalling 244.

Anchor / maximum multiplicity	2	3	4	5	6	7
Adjacent-repeat	72	55	31	14	4	1
Opposite-repeat	26	21	11	6	2	1
Total	98	76	42	20	6	2

For distinct collections a different choice of anchor reduces the count.

Lemma 5 (Shared two-colour corner)**.** *Among eleven adjacent-repeat cubes, two share a two-colour corner type. If a distinct collection contains such a pair and one member is chosen as anchor, the anchor's two two-colour positions cannot be Hall-deficient.*

Proof. Each adjacent-repeat cube has exactly two such types, *aab* and *aac*, with $b \neq c$. There are only $5 \cdot 4 = 20$ types *aab*. Eleven cubes contribute 22 incidences, so two cubes share a type. For the two characteristic positions of the chosen anchor, the anchor itself and the other cube are two distinct neighbours. Their neighbourhood therefore has size at least two. □

If an opposite-repeat variety is present, choose it as anchor; this gives 31 constraint orbits with every multiplicity at most one. Otherwise use Lemma 5 to choose an adjacent-repeat anchor. Omit the cover complementary to its two characteristic positions and set all opposite-repeat capacities to zero. Its stabiliser preserves these positions, so the omission is compatible with the orbit reduction. It removes one of the 83 adjacent-repeat constraint orbits, leaving 82 further cases.

Proposition 6. *Every solution-free fifteen-cube collection is carried by colour relabelling and, if needed, global reflection into one of the 357 exported root cases.*

Proof. The preceding arguments cover repeated and distinct collections. The finite catalogue, two-variety bounds and coverage of all anchor-cover orbits are checked from physical rotations by the accompanying audit. It regenerates rotations from quarter turns, separately from the signed-axis permutations used by the case generator, and checks every exported bound and every cover orbit. The resulting counts are

$$244 + 31 + 82 = 357.$$

The cases need not be disjoint; completeness of the cover is what is required. □

Each root specifies an anchor A of multiplicity m, coordinate bounds ℓ, u, a set S of non-anchor varieties and a budget b. The bounds fix $\ell_A = u_A = m$, with $\ell_i = 0$ for $i \neq A$. A vector $x \in \mathbb{Z}^{75}_{\geq 0}$ is *admissible* for this root if

$$\ell \leq x \leq u, \qquad \sum_{i=1}^{75} x_i = 15, \qquad x(S) \leq b, \tag{3}$$
$$\text{for every target } T, \text{ there is } \varnothing \neq D \subseteq \{1, \ldots, 8\} \text{ with } x(N_T(D)) \leq |D| - 1.$$

Thus admissibility includes failure of every target, in addition to the anchor constraints.

The cover can be reduced without further search. If $S' \subseteq S$ and $b' \geq b$, then $x(S) \leq b$ implies $x(S') \leq b'$ for nonnegative x. Together with coordinatewise inclusion of the root boxes, this identifies 49 cases contained in other cases, leaving 308. Write u for the root upper bounds. In 29 retained cases, $b + u(S^c) < 15$ already contradicts the prescribed total, so only 279 cases need further exclusion proofs. The exact map and retained case numbers are in `data/q5-root-reduction.json`; Lean checks all containments and arithmetic exclusions. The archive retains all 357 original cases.

Lemma 7 (Hall-budget projection)**.** *Suppose a root has anchor multiplicity m, budget $x(S) \leq b$, and total 15, with the anchor outside S. For $0 \leq s \leq b$, put $G = 15 - b + s$ and suppose $G \geq m$. Every admissible vector x has a vector $y \leq x$ with the same anchor multiplicity, total G, budget $y(S) \leq s$, and a Hall deficiency for every target. It obeys the original upper bounds, tightened to $\min(u_i, s)$ on S.*

Proof. First retain only $\min(x(S), s)$ copies in S. This removes at most $b - s$ copies and leaves the anchor untouched, so the remaining total is at least G. Delete further non-anchor copies until the total is G; this is possible because $G \geq m$. Every coordinate decreases, so all upper bounds and Hall deficiency inequalities are preserved. Nonnegativity and $y(S) \leq s$ give $y_i \leq s$ for $i \in S$. □

For our anchored covers, $b = 7 - m - |K|$, so

$$G = m + 8 + |K| + s.$$

Thus the reduced collection consists of the m fixed anchor copies and $8 + |K| + s$ other copies; in particular, $G \geq m$ is automatic. Apply the lemma after selecting and normalising the original root, retaining its bounds. No new anchor selection is required for the smaller collection. Of the 279 remaining roots, 277 admit an exclusion certificate with $s = 0$. The other two, original cases 68 and 69, use $s = 1$ and $G = 12$. Their $s = 0$, $G = 11$ problems actually have solution-free collections, supplied and checked by physical matching in the projected manifest. The projected totals range from 9 to 15. All 101 roots whose original budget was zero retain their original certificates.

5 Interval certificates

For a projected problem, admissibility is defined by (3) with tightened upper bounds and total G. In this section denote its projected budget by b. We certify that no admissible vector exists.

Write a box as $\ell \leq x \leq u$. For a cut $x(S) \leq r$, two necessary conditions for feasibility in the box are

$$\ell(S) \leq r, \qquad G \leq r + u(S^c). \tag{4}$$

The first follows from the lower bounds; the second follows from $G = x(S) + x(S^c)$. When they hold, put $\sigma = r - \ell(S)$ and $\gamma = r + u(S^c) - G$. Every satisfying vector obeys

$$x_i \leq \min(u_i, \ell_i + \sigma) \quad (i \in S), \qquad x_i \geq \max(\ell_i, u_i - \gamma) \quad (i \notin S). \tag{5}$$

For example, the upper bound follows by giving all other coordinates of S their lower bounds; the lower bound follows by giving all other coordinates of S^c their upper bounds.

For a target, retain its cuts satisfying (4). If none remain, no vector in the box can fail that target. Otherwise let I and U be the intersection and union of their neighbourhoods, and let σ_*, γ_* be the largest corresponding σ, γ. Then all vectors in the box that fail this target remain in the refined box

$$u_i' = \begin{cases} \min(u_i, \ell_i + \sigma_*) & i \in I, \\ u_i & i \notin I, \end{cases} \qquad \ell_i' = \begin{cases} \max(\ell_i, u_i - \gamma_*) & i \notin U, \\ \ell_i & i \in U. \end{cases} \tag{6}$$

Indeed, a failing vector satisfies one of the retained cuts, and its bounds in (5) imply (6). The root budget is treated as a family containing a single cut. The total $\sum_i x_i = G$ gives

$$\ell_i' = \max\left(\ell_i, G - \sum_{j \neq i} u_j\right), \qquad u_i' = \min\left(u_i, G - \sum_{j \neq i} \ell_j\right).$$

A total-bound leaf closes exactly when $\sum_i \ell_i > G$ or $\sum_i u_i < G$. A budget leaf closes when $\ell(S) > b$ or $b + u(S^c) < G$; a target leaf closes when none of its 255 cuts satisfies (4). The checker thus has 77 closing reasons: one total test, one budget test and 75 target tests. Refinements must tighten at least one bound and leave nonempty coordinate intervals; a split must satisfy $\ell_i \leq v < u_i$.

A certificate is a finite tree of refinements and integer splits $x_i \le v$ or $x_i \ge v+1$. Each leaf is excluded by total bounds or by a cut family with no surviving cut. Induction on this tree proves that its root contains no admissible vector. The separate C++ checker reads these instructions and verifies the deductions; it performs no search and does not use the search program's matching implementation.

Proposition 8 (Certificate replay)**.** *The separate checker accepts an exclusion certificate for each of the 279 projected problems.*

The projected archive contains 499,829 instructions, 95.69% fewer than the original archive. Both archives and the search computations are documented in the repository. Every archived header is checked against its exported root case. Eighty negative controls are rejected, including all 77 possible direct closing claims against a separate $G = 11$ fixture, not a $G = 15$ root problem. The witness satisfies its total, fixed anchor count, coordinate bounds and zero root budget, and fails all 75 targets by physical matching. The fixture is `data/q5-distinct-known-control.txt`; its regeneration and witness checks are in `scripts/check_five_colour_distinct.py`.

Proof of Theorem 1. Lemma 3 gives $N_5 \ge 15$. A solution-free collection of fifteen cubes would satisfy an exported root case by Proposition 6, and would supply a Hall deficiency for every target. Containment and the initial budget tests leave one of 279 roots. Lemma 7 supplies an admissible vector in its projected problem, which Proposition 8 and the sound interval deductions exclude. Hence $N_5 \le 15$. □

6 Frames and complete cubes

The larger-cube argument depends on how many positions a source can miss. The following form of Hall's theorem isolates the matching step.

Lemma 9 (Hall criterion with a matched subset)**.** *Consider L target positions and $N \ge L$ source copies. Suppose every source misses at most d positions. If a set P_0 of positions can be matched to distinct copies, and every position outside P_0 has at least d neighbours, then all L positions can be matched.*

Proof. For a position set $D \subseteq P_0$, the given matching supplies $|D|$ neighbours. If $D \not\subseteq P_0$ and $|D| \le d$, a position in $D \setminus P_0$ alone has at least d neighbours. If $|D| > d$, every source neighbours D, since none can miss all its positions. In each case Hall's inequality holds. □

Lemma 10 (Frame compatibility)**.** *Put $q = n - 2 \ge 1$. Every source variety fits at least $8q+2$ positions of every five-colour n-frame target, and at least $9q+2$ if the target is opposite-repeat. An adjacent-repeat source lacks exactly one bichromatic edge type. An opposite-repeat source lacks exactly two, which are disjoint; these two missing pairs determine its variety uniquely.*

Proof. A missing bichromatic pair must occupy opposite singleton faces: if a colour is repeated, a face of the other colour cannot be opposite both copies. An adjacent-repeat variety has one opposite pair of singleton faces, and an opposite-repeat variety has two. Two disjoint missing pairs specify two opposite face pairs, leaving the fifth colour on the remaining opposite faces, which determines a unique variety.

Write c, e for the numbers of target corners and geometric edges fitted by the source. Whenever either variety is opposite-repeat, $c \ge 2$. To see this, let that variety repeat r, with singleton opposite pairs $\{a, b\}$ and $\{u, v\}$. It contains both cyclic orientations of each rxy with $x \in \{a, b\}$ and $y \in \{u, v\}$. In the other variety, if an r-face has no adjacent r-face, its four surrounding faces contain at least three distinct colours. Their cyclic order crosses between the two pairs at least twice, giving two shared corner types. If the r-faces are adjacent, the other neighbours of each form a path of three distinct colours. Each path crosses between the pairs and gives

a shared corner; these corners have different types. Every five-colour variety has eight distinct cyclic corner types: the only repeated colour multisets occur at the two ends of an edge, with opposite cyclic orientations. Shared corner counts are therefore symmetric in the source and target.

A bichromatic target edge type occurs at most twice, and a doubled type contains the target's repeated colour. Two disjoint types therefore account for at most three target edges. There is at most one monochromatic target edge. These observations give the following bounds; A and O denote adjacent-repeat and opposite-repeat varieties.

Target	Source	c	e	$c+e$
O	O	≥ 2	≥ 9	≥ 11
O	A	≥ 2	≥ 10	≥ 12
A	O	≥ 2	≥ 8	≥ 10
A	A	≥ 0	≥ 9	≥ 10

Only the final entry needs an orientation argument. If an adjacent-repeat source fits exactly nine edges of an adjacent-repeat target, it misses the monochromatic edge and a doubled bichromatic type. Relabel colours and rotate to put the target at $(0, 1, 0, 2, 3, 4)$ and the missing type at 03. The source's 0- and 3-faces are opposite; its surrounding colours are $1, 2, 4$, one repeated on adjacent faces. Suppress this adjacent repetition. One of the two cyclic orders gives the shared target corner 123; the other gives 024 and 041. Thus $c \geq 1$ when $e = 9$.

Each geometric edge contributes q positions. Since $e \geq 8$ and $c + e \geq 10$,

$$c + qe = (c + e) + (q - 1)e \geq 8q + 2.$$

For an opposite-repeat target, $e \geq 9$ and $c + e \geq 11$ instead give $9q + 2$. □

For a bichromatic pair ab, let $\mathcal{H}_{ab}$ be the family of varieties lacking that edge type. In every such cube, a and b are opposite singleton faces. The four faces around this axis use the other three colours, one of them twice. There are nine varieties in $\mathcal{H}_{ab}$: three opposite-repeat varieties O_r, and six adjacent-repeat varieties A_r^+ and A_r^-, where r is the repeated colour. The signs distinguish the two cyclic orientations of the three distinct ring colours. All three A_r^+ varieties share the same six tricoloured corner types; all three A_r^- varieties share the reverse six types.

Lemma 11 (A fixed missing pair)**.** *Fix a bichromatic pair ab and eleven cubes from $\mathcal{H}_{ab}$.*

(i) They have a corner solution whose target belongs to $\mathcal{H}_{ab}$.

(ii) If at least one of the eleven cubes is opposite-repeat, they can match the eight corners and the monochromatic edge position, when present, of a 3-frame with target in $\mathcal{H}_{ab}$.

(iii) After adjoining any one five-colour cube, they can match those positions for a target in $\mathcal{H}_{ab}$.

Moreover, every variety in $\mathcal{H}_{ab}$ shares at least four corner types with every opposite-repeat variety in this family.

Proof. Let o_r, p_r, m_r count the varieties O_r, A_r^+, A_r^-, respectively, and write $O = \sum_r o_r$, $P = \sum_r p_r$, and $M = \sum_r m_r$. Thus $O + P + M = 11$.

The eight corners of an opposite-repeat target O_r form four pairs arranged as a 2×2 array. The rows correspond to the two unordered ring-colour pairs involving r; the columns correspond to their positive and negative cyclic orientations. Each cell represents two corner positions, one at each end of the ab axis. A source O_r fits all four cells, the other two opposite-repeat varieties each fit one row, and the positive and negative adjacent-repeat varieties fit the corresponding columns. This proves the final assertion of the lemma.

For example, in $\mathcal{H}_{01}$ take $O_2 = (0, 1, 2, 2, 3, 4)$ and choose $+$ to include the corner 023. Its four cells are

	$+$	$-$
23	023, 132	032, 123
24	042, 124	024, 142

Here cyclic rotations identify corner labels. This also makes explicit why both positions in a cell have the same neighbourhood.

Suppose that

$$O + \min\{P, M\} \geq 4,$$

and choose r with o_r maximal. We verify Hall's inequalities for the array of target O_r. It suffices to consider unions of complete cells, since the two positions in a cell have identical neighbourhoods. Each single cell has at least

$$O/2 + \min\{P, M\} \geq 2$$

neighbours: the omitted opposite-repeat count is at most $O/2$. Each row has at least $11 - O/2 \geq 11/2 > 4$ neighbours, and each column has at least $O + \min\{P, M\} \geq 4$. Every other nonempty union of cells, apart from single cells, rows, and columns, has all eleven cubes as neighbours. Hall's theorem gives a corner solution on O_r. It has no monochromatic edge, so this proves all three parts in this case.

Otherwise, exchange the signs if necessary so that $O + M \leq 3$. Then $P \geq 8$, and some A_r^+ occurs at least three times. Two copies fill its two repeated-colour corners, and six other positive copies fill its six tricoloured corners. This proves (i).

If $P \geq 9$, three copies of A_r^+ fill its two repeated-colour corners and its monochromatic edge position; six other positive copies fill the tricoloured corners. If $P = 8$ and $O > 0$, use three copies of A_r^+ in the same way. An opposite-repeat source in $\mathcal{H}_{ab}$ shares four tricoloured corners with this target, so it fills one of them, and the five other positive copies fill the remaining five. These arguments prove (ii), and also prove (iii) unless

$$O = 0, \qquad P = 8, \qquad M = 3.$$

In this last case, let W be the additional cube and again choose $A = A_r^+$ with $p_r \geq 3$. If W fits any corner or the monochromatic edge position of A, place it there. The eight positive copies fill the remaining eight positions: at most three of them are needed at the repeated-colour corners and monochromatic edge, and all positive copies fit every tricoloured corner of A.

If W fits none of those positions, it shares no corner with A. Take the opposite-repeat target $O_r \in \mathcal{H}_{ab}$. Its corners split into four positive positions, all present on A, and four negative positions. By Lemma 10, W shares at least two corners with O_r; these must be negative positions. Place W at one of them, the three negative copies at the other three, and four positive copies at the four positive positions. This gives an opposite-repeat corner solution and completes (iii). □

Theorem 12 (Sharp five-colour frame numbers)**.**

$$F_5(n) = 12n - 16 \qquad \textit{for every } n \geq 3.$$

Proof. The number of frame positions gives the lower bound. For the upper bound, fix $q = n - 2 \geq 1$ and an arbitrary collection of $N = L = 12q + 8$ source cubes. Call a bichromatic pair *rare* if at most $4q + 5$ of these cubes possess that edge type. Each rare pair is therefore missing from at least $8q + 3$ cubes. Two pairs sharing a colour cannot both be missing from one five-colour cube. Since $2(8q + 3) > 12q + 8$, any two rare pairs must be disjoint. There are at most two rare pairs in a five-colour palette.

Whenever a target avoids all rare pairs and its corners and monochromatic edge positions have a matching, Lemma 9, with $d = 4q + 6$, completes its frame. Every remaining position has at least d neighbours, and Lemma 10 bounds the positions missed by a source.

No rare pair. For each colour r, reserve $\min\{q, A_r\}$ cubes, where A_r is the number of sources with adjacent r-faces. These five reservations are disjoint, and leave at least $7q + 8 \geq 15$ cubes. Apply the corner theorem to the unreserved cubes. If its target has a monochromatic edge rr, a source with adjacent r-faces survived the reservation; therefore the r-reserve has its full size q. It fills all positions on that edge. The target thus has the required partial matching, and Lemma 9 completes the frame.

One rare pair ab. There are at least $8q + 3$ cubes in $\mathcal{H}_{ab}$. If $q \geq 2$, reserve up to q adjacent-repeat cubes of each repeated colour within this family, using the same rule as above. Only three colours can be repeated, so at least

$$(8q + 3) - 3q = 5q + 3 \geq 11$$

cubes remain. Apply Lemma 11(i) to eleven of them. The target belongs to $\mathcal{H}_{ab}$, and any monochromatic edge is filled from its full reserve of q cubes. If $q = 1$, instead choose eleven cubes in $\mathcal{H}_{ab}$ and any one further source copy; part (iii) of that lemma directly gives the corner and monochromatic edge matching. In either case the target avoids the only rare pair, so Lemma 9 completes the frame.

Two rare pairs e, f. The pairs are disjoint. A cube missing both has the fifth colour repeated on opposite faces and e, f as its singleton opposite pairs. These data determine one opposite-repeat variety V. Its multiplicity v satisfies

$$v \geq N - 2(4q + 5) = 4q - 2.$$

If $q \geq 2$, then $v \geq 6$. When $v \geq 8$, use eight copies for the corners of V. Otherwise v is six or seven; choose two other source copies and place them at distinct corners of V, which is possible because each shares at least two such corners. Six copies of V fill the remaining corners. The target has no monochromatic edge and avoids both rare pairs, so Lemma 9 applies.

It remains to treat $q = 1$. If $v \geq 8$, use eight copies for the corners of V as before. If $4 \leq v \leq 7$, the family $\mathcal{H}_e$ contains at least eleven cubes and hence at least four copies not of variety V. Each shares at least four corners with V, by Lemma 11. Place four of these copies greedily at distinct corners and fill the other four with V. Again the target avoids both rare pairs and has no monochromatic edge.

Finally $v \in \{2, 3\}$, since $v \geq 4q - 2 = 2$. Let $s_e, s_f \leq 9$ count the cubes possessing the two rare pairs. Since

$$v \geq 20 - s_e - s_f,$$

both s_e and s_f are at least eight. Choose eleven cubes from $\mathcal{H}_e$, including a copy of V, and apply Lemma 11(ii). This gives a target $T \in \mathcal{H}_e$ with its corners and monochromatic edge position matched. A copy of V fits at least thirteen positions of this 3-frame: it shares at least four corners with T, and misses at most the two geometric edges of type f and the monochromatic edge.

We verify Hall's inequalities for all twenty positions of T, using all twenty original source copies. Sets with no bichromatic edge position are covered by the partial matching. Sets containing an edge position of a non-rare type are covered by the earlier Hall argument. In any remaining set D, every bichromatic edge position has type f, since T avoids e. There are at least s_f neighbours, so assume $|D| > s_f \geq 8$. Since V fits at least thirteen positions, it misses at most seven; hence every copy of V is also a neighbour of D. None of these copies possesses f, so they are distinct from the s_f neighbours already counted. Thus

$$|N(D)| \geq s_f + v \geq 20 - s_e \geq 11.$$

If any source copy is not a neighbour, the compatibility bound gives $|D| \leq 10$; otherwise all twenty copies are neighbours. Hall's inequality holds in either case.

All cases yield a matching of the entire frame. The final matching is allowed to rearrange the preliminary corner placements and to use any of the original source copies; the preliminary construction specifies the target and verifies the required subset of Hall's inequalities. □

Theorem 13. *For every integer $n \geq 2$,*

$$G_5(n) = \max\{n^3, 15\}.$$

In particular, every collection of n^3 five-colour cubes can all be assembled into a complete n-cube for every $n \geq 3$.

Proof. For $n = 2$, a complete cube is its corner frame, so Theorem 1 gives $G_5(2) = 15$. For $n \geq 3$, the piece count requires $G_5(n) \geq n^3$. Conversely, $n^3 \geq 12n - 16$, since the difference counts face-interior and interior positions. Theorem 12 therefore supplies a frame from any n^3 sources. Each remaining face-interior position prescribes just one colour, which every unused cube has, and interior positions prescribe none. All remaining cubes can fill these positions. □

Thus any twenty cubes form a 3-frame, and any 27 cubes form a complete $3 \times 3 \times 3$ cube, without spares in either case.

7 Collections with few varieties

Let $M(k)$ be the least number of cubes guaranteeing a corner solution when the collection contains at most k varieties.

Lemma 14. *Two five-colour varieties with disjoint corner sets can build an opposite-repeat target from four copies of each.*

Proof. This is a small finite geometric check. Of the $\binom{75}{2}$ unordered pairs, 330 have disjoint corner sets. Up to colour permutation and exchange of the sources, they have the following five representatives. The first source is 010234 throughout; tuples use the face order of Section 2. Each listed target has four corners supplied by each source.

Second source	Target
011243	001234
012324	012234
012334	012234
012434	012234
030412	001234

The accompanying program `src/five_colour_extensions.cpp` regenerates the varieties and checks the orbit coverage and all five witnesses. □

Theorem 15. *For every integer $k \geq 1$,*

$$M(k) = \begin{cases} 8 & k = 1, \\ 13 & k = 2, \\ 15 & k \geq 3. \end{cases}$$

Consequently, three is the minimum number of varieties in a fourteen-cube collection without a solution.

Proof. Eight identical copies build their own variety, and fewer than eight pieces cannot form a corner solution. For two varieties, consider thirteen copies. If either variety occurs at least eight times, it builds itself. Otherwise the multiplicities are seven and six. If the varieties share a corner type, use a minority copy at that corner of the majority target and seven majority copies elsewhere. If their corner sets are disjoint, apply Lemma 14.

For the lower bound take six copies each of

$$P = (0, 1, 0, 2, 3, 4), \qquad Q = (0, 1, 1, 3, 2, 4).$$

Their corner sets share just 041. Inspection of the 75 varieties shows that only P, Q have all corner types in their union. Each has seven private positions, so neither can be built with six copies. The same geometric checker verifies these assertions. Thus $M(2) = 13$. For $k \geq 3$, Theorem 1 supplies the upper bound, and the collection in Lemma 3 supplies the lower bound. □

Remark (A family of fourteen-cube obstructions). Keep A, C from Section 3, and put

$$B_1 = (0, 1, 2, 3, 2, 4), \quad B_2 = (0, 1, 2, 3, 3, 4), \quad B_3 = (0, 1, 2, 4, 3, 4).$$

All 21 collections

$$4A + 5C + x_1B_1 + x_2B_2 + x_3B_3, \qquad x_1 + x_2 + x_3 = 5, \quad x_i \in \mathbb{Z}_{\geq 0},$$

are unsolvable. The B_i share the six trichromatic corner types $023, 034, 042, 124, 132, 143$. The combined corner union allows only A, C, B_1, B_2, B_3 as targets, as a check of the catalogue confirms. Target A needs five A-copies, C needs six C-copies, and any B_i needs six copies from the B-group. The respective supplies are four, five and five, regardless of their distribution within the B-group. The original example is $x_1 = 5, x_2 = x_3 = 0$.

8 Formalisation and reproducibility

The development uses Lean 4 and mathlib [7, 8], both at `v4.35.0`. It proves the physical theorems, including the connection between cube arrangements and the finite certificate problems.

Lean proves completeness and uniqueness of the 75 varieties and checks all 45,000 corner-compatibility bits using the shared corner–rotation theorem. The lower bound uses 75 deficient corner sets: 72 singletons and three sets of sizes 5, 6 and 6. Its theorem is `FiveColour.Physical.lower_bound` in `FiveColour/LowerBound.lean`. This replaces the elementary target classification in Lemma 3; catalogue completeness and rotation transport extend the result to every physical target.

The upper-bound formalisation follows the Hall reduction, symmetry transport, anchor selection and shared-corner argument. Finite checks verify capacities, cover representatives and the reduction from 357 roots to 279. The deletion step is `FiveColour.Physical.root_projection` in `FiveColour/RootProjection.lean`. The bridge to the archive, in `FiveColour/ProjectedBridge.lean`, identifies the full admissibility predicate: coordinate bounds, total, budget and all Hall-cut families.

For efficient replay, Lean proves soundness of a packed implementation of the interval checker. For multiplicities $0 \leq x_i \leq 7$, index the varieties by $0, \ldots, 74$ and put

$$P(x) = \sum_{i=0}^{74} x_i B^i, \qquad W(S) = \sum_{i \in S} B^{74-i}, \qquad B = 1024.$$

Then

$$x(S) = \left\lfloor \frac{P(x)W(S)}{B^{74}} \right\rfloor \bmod B.$$

Every coefficient in the product is at most $75 \cdot 7 < 1024$, so no carry can affect this coefficient extraction. Lean proves the identity, decoding, coordinate updates and equivalence with the integer inequalities, and checks all 19,125 neighbourhood weights.

The 279 certificates are split into 4,786 chunks of at most 128 instructions. Composition requires exact boundary-vector agreement and a proof for every referenced box; the exporter is untrusted. All chunks and their final assembly have passed kernel checking. The case mapping and certificate hashes are in `data/q5-projected-root-manifest.json`. The generated module `FiveColour/ProjectedChunks/Threshold.lean` proves the unconditional theorems `threshold15` and `sharpThreshold15`. Both lie in namespace `FiveColour.Packed.ProjectedChunks`.

The larger-cube argument uses general Hall and reserve lemmas.

`FiveColour/RingGeometry.lean` checks the nine-variety missing-pair patterns against physical compatibility. For each target, ten inequalities (including the empty-set inequality) cover every Hall set; `FiveColour/RingWeights.lean` verifies this finite reduction. `FiveColour/RingEleven.lean` then proves the three eleven-cube lemmas by arithmetic on multiplicities and a matching with one extra source, following Lemma 11. `FiveColour/FixedPair.lean` transfers these statements to physical cubes. This does not enumerate eleven-cube collections. A supplementary C++ audit checks all 75,582 multiplicity vectors in one missing-pair family and all 1,800 extensions of the 24 collections without a core matching.

Lean also checks the general geometric bounds for all 5,625 source–target pairs, the common missing-pair bounds, and uniqueness of a variety lacking two pairs. Edge bits are checked against face adjacency; corner counts reuse the verified incidence table. The theorem `FiveColour.Physical.sharp_frame_of_threshold` in `FiveColour/SharpFrames.lean` deduces the sharp frame bound for every $q \geq 1$ from `Threshold15`. Frame completion in `FiveColour/FullCube.lean` adapts the general construction from [4] without using its threshold. The small-support proofs check pair witnesses directly, without reducing them to five colour orbits, and use deficient corner sets for the two obstruction constructions. They are in `FiveColour/SmallSupport.lean` and `FiveColour/MixedObstructions.lean`.

`Solution.lean` exports the unconditional theorems. In namespace `FiveColour.Puzzle`, these include `frame20`, `frame_threshold_all`, `full_cube`, `complete_cube_threshold` and `support_threshold`.

The dependency audits cover 114 declarations listed in `Audit.lean`, nine extension statements listed in `ExtensionsAudit.lean`, and the final replay theorems. Their dependencies are confined to `propext`, `Classical.choice` and `Quot.sound`. No `native_decide` is used. Reproduce the proof and audits with:

```
make check WORKERS=2
make projected-replay-data WORKERS=8
make projected-replay WORKERS=8
make check-extensions
```

The module guide is `FiveColour/README.md`.

A Mathlib-only comparator challenge states the corner problem geometrically. Its equivalence with the development's model is proved by `FiveColour.Puzzle.hasSolution_iff` in `FiveColour/ComparatorBridge.lean`. The solution module applies the completed threshold theorem. A full external comparator pass remains unconfirmed; it is separate from the completed Lean proof.

9 Remaining questions

The sharp frame and complete-cube thresholds leave the structure of extremal corner obstructions to be determined. Can all fourteen-cube obstructions be classified? The support bound and the family in Section 7 give initial constraints and examples.

AI disclosure and review status

GPT-6-Astra (OpenAI), used through Codex, developed the arguments, search and verification programs, Lean formalisation and manuscript under the human author's direction. The author initiated and directed the research. Author review and submission approval of this working manuscript are pending. The inherited geometry has its own provenance in the pinned dependency. Automated checks and AI review are not independent human peer review.

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